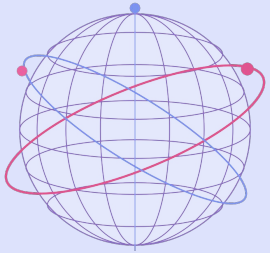
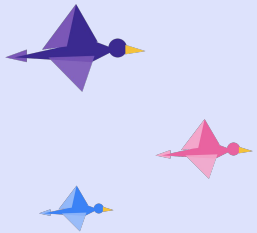
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details





Participant Details

PROJECT TITLE
Q-Resilience AI · Brand: Q-Silience

TRACK
Quantum + AI & ML

TEAM NAME
Ctrl_AI

TEAM LEADER
Dhyan P

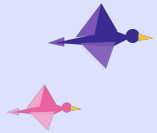
EMAIL / CONTACT
dhyanp1355@gmail.com

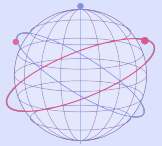
TEAM MEMBERS
Dhyan P · Sangamesh Athani · Harsha Hiregoudar

INSTITUTION / ORGANISATION
Jain College of Engineering and Research, Belagavi

COUNTRY
India

Problem Statement





THE PROBLEM

Microgrids juggle demand, solar, battery, voltage and frequency at the same time.
Single-threshold alarms watch one signal at a time and can miss unusual combinations, such as a demand surge while solar output drops and voltage sags.

RESEARCH QUESTION

Can a quantum-kernel classifier (QSVC) label simulated microgrid records as normal or anomalous, and how does it compare with a classical SVC on the same held-out data?

WHY IT MATTERS

1. Demand surges, solar drops, low battery and voltage or frequency drift often combine to threaten stable operation.
2. Q-Resilience, our dashboard, makes model outputs and sensor patterns easy for an operator to inspect.
3. A fair quantum-vs-classical comparison on identical data turns a hype question into a measured result.

Objectives & Novelty

OBJECTIVES

1. Generate or upload simulated microgrid records (5 channels generated; models use 4)
2. Train a Qiskit quantum-kernel QSVC and a scikit-learn SVC on the same split
3. Evaluate both on held-out data: accuracy, precision, recall, F1, runtime
4. Present predictions, metrics and scenarios in the Q-Resilience dashboard

NOVELTY

- **Measurable experiment.** One protocol scores quantum and classical models on the same held-out set.
- **Scenario Simulation Lab.** Scenario presets are sent to the backend; labels come from the models, not hardcoded.
- **Transparent by design.** Readiness indicators, error states, simulation labels and stated limits.

Solution & Quantum Advantage

Sensor data › Preprocessing › **QSVC ‖ SVC** › Held-out evaluation › Dashboard

THE SOLUTION

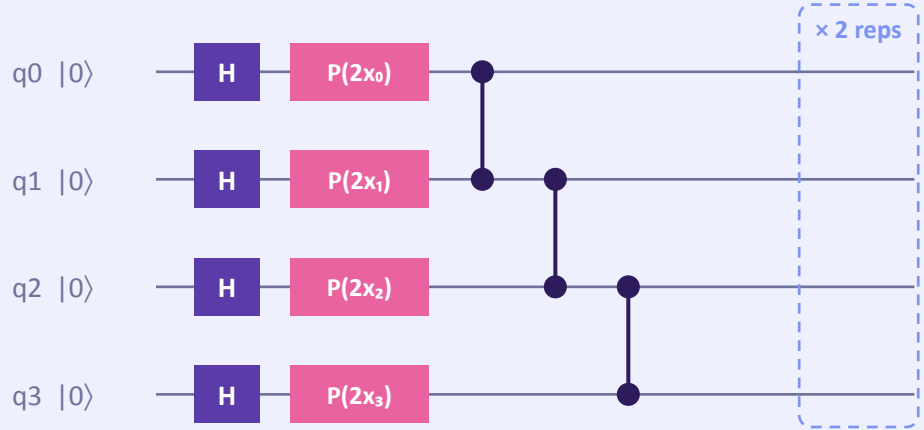
- ZZFeatureMap encodes scaled sensor features into quantum states
- A fidelity-based quantum kernel feeds a Qiskit QSVC
- A scikit-learn SVC runs as the baseline on identical data
- Dashboard shows the prediction and which model produced it

QUANTUM ADVANTAGE: OUR POSITION

- We do not claim quantum advantage
- We measure whether a quantum kernel is competitive: accuracy, F1 and runtime on the same held-out data
- Backend run: classical SVC led (F1 1.00 vs 0.86, 15 test records). Simulator only, so no hardware claim

Algorithm & Circuit Design

Illustrative ZZFeatureMap schematic: 4 features → 4 qubits



H = Hadamard · P = phase rotation by feature x_i · ●—● = ZZ entangling interaction (linear)

KEY DETAILS

- **Encoding:** 4 features scaled to [0, π]; ZZFeatureMap, reps = 2
- **Kernel:** Fidelity QuantumKernel (noise-free simulation)
- **Classifier:** Qiskit ML QSVC; statevector-kernel SVC as fallback
- **Execution:** classical simulation only; no hardware runs
- **Circuit:** 4 qubits, linear entanglement, depth 19, 34 gates (GET /api/circuit)

Implementation & Tech Stack

FRONTEND
React 19 · TypeScript · Vite · Recharts

BACKEND API
Python · FastAPI · Uvicorn

QUANTUM
Qiskit + Qiskit Machine Learning (QSVC, simulator)

CLASSICAL ML
scikit-learn SVC, scaling, metrics

DATA
NumPy · pandas · synthetic generator

HOSTING
Vercel (frontend) · Render Web Service (API)

API ENDPOINTS

`GET /health`
status + model readiness

`GET /api/circuit`
ZZFeatureMap depth, gates

`POST /api/predict`
one record, both models

`GET /api/benchmark`
metrics (+ POST /api/evaluate)

`GET /api/scenarios`
preset scenarios

Stack and endpoints checked against backend/main.py and frontend/package.json.

Results & Analysis

Quantum vs classical on the held-out test set (15 records)

Model	Accuracy	Precision	Recall	F1	Inference
QSVC (quantum)	0.867	0.857	0.857	0.857	≈19ms
SVC (classical)	1.000	1.000	1.000	1.000	≈84.8ms

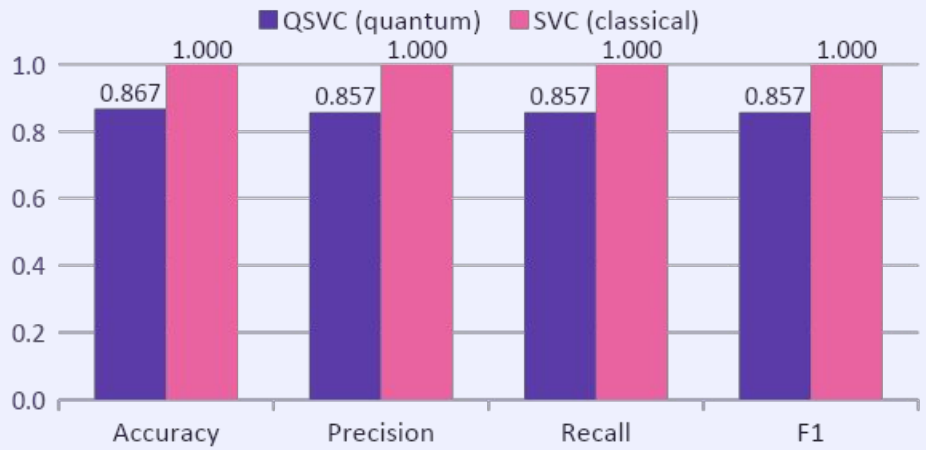
Backend run · synthetic data. SVC 100% vs QSVC 86.7% (13/15): QSVC had 1 missed anomaly, 1 false alarm. *Timed on a local re-run; README logs ~16.8 s vs 0.29 ms on Render. No quantum advantage claimed.

EVALUATION PROTOCOL

- Same split and scaling for both models; scaler fit on training data only
- Fixed settings (reps = 2, scale [0, π], C = 1); no tuning on the test set
- Report accuracy, precision, recall, F1, confusion matrix, runtime
- Backend default: seed 42, 60 records (45 / 15), 25% stratified test, 27 anomalous
- Claim superiority only if the measurements show it

Where Each Model Fails

Held-out metrics, QSVC vs SVC (15 test records)

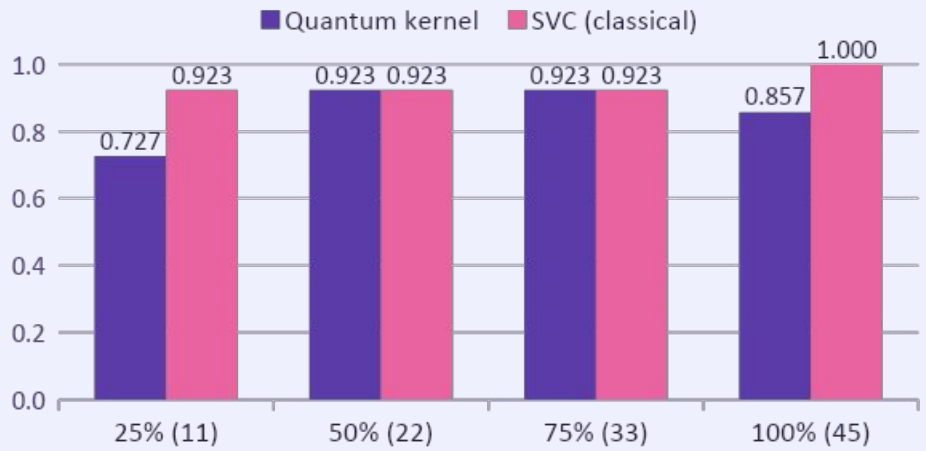


ERRORS ON 15 TEST RECORDS

- QSVC: caught 6 of 7 anomalies, 1 false alarm
- SVC: caught 7 of 7 anomalies, 0 false alarms
- Recall gap = 1 missed anomaly; in a grid a miss costs more than a false alarm
- Tiny test set: one error shifts recall by 14 points

Insight: Sample Efficiency

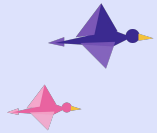
Test F1 vs training-set size (same 15 held-out records)

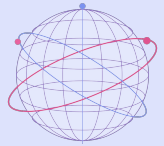


WHAT WE LEARNED

- QSVC kernel trailed or tied SVC at every training size
- Best quantum F1 (0.92) at 22–33 records; 0.86 at the full 45
- No sample-efficiency advantage observed
- Quantum bars: statevector fidelity kernel + SVC, not the QSVC object
- 15 test records: indicative, not final

Impact & Real-World Applications





1

Microgrid monitoring

Surfaces unusual combinations of demand, solar, battery, voltage and frequency that single-threshold checks can miss. Predictions are model outputs, not confirmed faults.

2

Measurable QML research

A fair comparison under one protocol. On our run the classical SVC led, and we report that openly.

3

Reusable demo pipeline

An end-to-end template covering data, preprocessing, QSVC, SVC, API and dashboard for teaching or extension.

Prototype on simulated data: not for controlling grid equipment or safety-critical decisions.

Challenges & Future Scope

CHALLENGES

- Synthetic labels may encode the generation rules, so high scores do not prove real-world performance
- Simulator results are not the same as running on quantum hardware
- QSVC trailed SVC (86.7% vs 100% accuracy) and was far slower; test set is only 15 records
- Free hosting means cold starts and limits; pin and test Python versions

FUTURE SCOPE

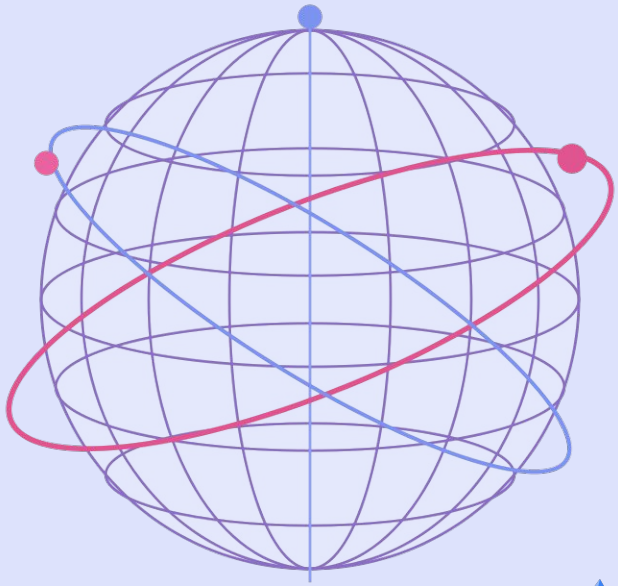
- Validate on real or public microgrid datasets, with repeated seeds and larger test sets
- Run on quantum hardware via Qiskit Runtime (not implemented yet; simulation only)
- Run the Circuit Lab, Investigation and Digital Twin views on backend results, not only local demo
- Add event history / export

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.



GITHUB REPOSITORY
https://github.com/vtu-qiskit-fall-fest-2026/vtu-hc-ctrl-ai.git

DEMO VIDEO LINK
https://qresilience.vercel.app/